

Nilanjan Sarkar

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Research Focus: pretraining optimization for large language models, LLM safety evaluation, mechanistic interpretability, recommender systems, and ML systems.

Education

IIIT Hyderabad

MS by Research, Computer Science and Engineering

Research Focus: Pretraining Optimization for Large Language Models

Hyderabad, India

Aug 2025 – Expected 2027

- Advisor: **Prof. Anil Kumar Nelakanti**. Researching CPU-side data-selection methods that use cheap gradient-proxy signals to select useful next-token training chunks under fixed token and compute budgets.
- Studying LSH/ANN-style retrieval over lightweight representations as an alternative to expensive per-example gradients or exhaustive full-corpus model scoring.

CGPA: 8/10

Heritage Institute of Technology

B.Tech., Computer Science and Engineering

CGPA: 7.31/10

Kolkata, India

Aug 2011 – Jul 2015

Research Experience

Data-Efficient Language Model Pretraining

MS Research under Prof. Anil Kumar Nelakanti

IIIT Hyderabad

Aug 2025 – Present

- Researching pretraining optimization for LLMs through CPU-side data-selection methods, where lightweight samplers choose useful training chunks under fixed token and compute budgets.
- Designing LSH-table-based candidate retrieval over gradient-proxy signatures, using low-cost signals from LM-head gradients, hidden states, and loss statistics to approximate high-utility training examples without full per-example gradient computation.
- Evaluating hash-based sampling strategies against random sampling, loss-banded sampling, hard mining, and bucket-balanced sampling in controlled small-LM pretraining experiments.
- Building reproducible pretraining infrastructure with cached tokenization, sbatch-based cluster execution, W&B experiment tracking, and post-hoc validation analysis.

Probing Devanagari as a Candidate Latent Bridge in Multilingual Indic Language Models

Independent NLP / Model-Internals Study

IIIT Hyderabad

Jan 2026 – May 2026

- Designed a RomanLens-style probing study to test whether multilingual Indic LMs use Roman, Devanagari, or Bengali-script intermediate representations during Indic-to-Indic translation, repetition, and cloze tasks.
- Built candidate-token construction and layerwise logit-lens analysis for native-script targets, Roman/Devanagari transliterations, bridge-hit rates, bridge mass, tokenizer fragmentation, and candidate-coverage controls.

Low-Rank Roads to Misalignment

Submitted Research – ICML Workshop

IIIT Hyderabad

Jan 2026 – ongoing

- Investigated whether emergent-misalignment directions are geometrically conserved across LoRA and full fine-tuning regimes in Qwen2.5-14B.
- Built a mechanistic evaluation pipeline with residual-stream mean-difference vectors, activation steering, projection ablation, same-norm random controls, and layerwise cosine analysis.
- Used calibrated open-source LLM-based behavioral judging to compare misalignment behavior against geometric interventions and random-control baselines.

University of Surrey

Research Assistant (Remote) under Prof. Nishanth Sastry

Guildford, UK

Jul 2023 – Jan 2024

- Built a Cookiebot-backed pipeline to capture and replay GDPR cookie-consent settings, studying whether stronger privacy defaults from GDPR regions can be carried into non-GDPR countries.
- Trained fastText classifiers on cookie-banner text and consent metadata to identify cookie categories and recommend privacy-preserving settings when local regulation is weaker.

Amazon

Applied Scientist Intern

(Done under online part-time M.Engg in CS program, 2020–22, BITS Pilani Hyderabad)

Bangalore, India

Jan 2022 – Jun 2022

- Built session-based refinement recommendation models over **1M+ products** and **200M+ customer interactions**, representing products as attribute-value dictionaries rather than raw ASIN IDs.
- Developed heuristic and fastText baselines, then co-developed **BERT4Ref** and **BERT4Ref+** with bidirectional self-attention and **whole-ASIN masking** to predict refinement attribute-values from browsing sessions.
- Evaluated Precision@K, Recall@K, MRR, coverage, popularity bias, and OPS; improved the heuristic baseline by **+545 bps at P@1** and **+503 bps at P@5**, with top-1 and top-5 recommendations appearing in **6.56%** and **37.55%** of purchase sessions.

- Co-authored and presented the work at the Amazon Machine Learning Conference (AMLC 2022); later publication work extended the refinement-recommendation findings.

Industry Experience

Digipay Software (Stealth Startup)

Senior Software Engineer

India

Jan 2024 – Jul 2025

- Owned backend and deployment for an insurance-brokering platform, covering partner integrations, policy workflows, payments, release, and monitoring.
- Built Kotlin/Spring Boot services with Kafka, Cassandra, MySQL, Docker, Jenkins, Argo CD, and Kibana; integrated four external partners across development, testing, and production rollout.

WESEE Project – Indian Navy Collaboration

Senior Research Engineer

India

Mar 2023 – Dec 2023

- Worked on an automatic MISRA code-compliance framework for critical defense software, using LLVM/Clang tooling to inspect C/C++ codebases and flag unsafe or non-compliant patterns.
- Contributed static-analysis checks around maintainability, reliability, and security-oriented engineering standards for high-assurance code.

Digital Green

Software Engineer II

Bangalore, India

Nov 2022 – Mar 2023

- Built Python/Django backend services and REST APIs for FarmStack data-exchange workflows, including secure-transfer and partner-integration components for agricultural data pipelines.
- Contributed to reliable data-sharing workflows across partner systems, with emphasis on access control, API correctness, and operational handoff.

Infosys

Senior Systems Engineer / Part-time Consultant

Hyderabad, India

Aug 2015 – May 2019; Feb 2020 – Dec 2021

- Built Java/Spring backend modules for banking and e-commerce clients, including JPA/Hibernate data layers, OAuth flows, PayPal Checkout integration, and JUnit/Mockito test coverage.
- Improved application efficiency by **34%** through multithreading and concurrency optimizations.

Selected Systems Projects

MP-QUIC File Transfer - Built a multipath QUIC file-transfer system and measured **135%** faster transfer than single-path QUIC under Mininet emulation.

Spectre Attack Proof-of-Concept - Implemented a cache-timing proof-of-concept for the Spectre side-channel attack.

Fuzzy Inference for Heart-Failure Mortality - Used genetic algorithms for feature selection, decision trees for rule selection, and MATLAB fuzzy inference on clinical-record data.

Skills

Languages: Python, Java, Kotlin, C++, C, SQL **ML / Research:** PyTorch, AWS SageMaker, NLP, Recommender Systems, LLM Pretraining, LLM Safety Evaluation

Backend / Systems: Spring Boot, Kafka, Cassandra, MySQL, Docker, Jenkins, Argo CD, Kibana, Django, REST APIs, LLVM, Clang

Coursework: Operating Systems, Computer Networks, Data Structures and Algorithms, Data Mining